

Isaac Mubiru

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PROFESSIONAL SUMMARY

Senior Backend Engineer with 4+ years building cloud-native infrastructure and real-time data systems. Led zero-downtime migration of 20+ production applications from AWS/Parse to GCP/Firebase, cutting infrastructure costs by \$5,000/month. Experienced in containerized workloads (Docker, Kubernetes), event-driven architectures (Apache Kafka, Google Pub/Sub), and cross-cloud migrations. Skilled at designing scalable backend systems with strong focus on reliability, cost efficiency, and observability.

CORE COMPETENCIES

Cloud Migration & Infrastructure

AWS & GCP Platforms

Docker & Kubernetes

Event-Driven Architecture

Apache Kafka & Pub/Sub

Real-Time Data Pipelines

Microservices Design

Cost Optimization

WORK EXPERIENCE

MTailor

September 2022 – Present

Software Engineer · US Remote

- Led end-to-end **cloud migration** of 20+ production applications from Parse/MongoDB on AWS to Firebase/GCP with **zero downtime**, reporting directly to the CTO
- Designed and implemented **real-time two-way data sync** between MongoDB and Firestore using Node.js and **Google Pub/Sub** for event-driven message processing
- Automated **cloud storage migration from AWS S3 to Google Cloud Storage** using Python, ensuring data integrity across all applications
- Achieved **\$5,000/month infrastructure cost savings** by migrating all services off AWS to optimized GCP architecture
- Built 3D visualization feature for customers using video overlay and ffmpeg library, increasing buyer conversion
- Implemented Express Shipping feature generating additional \$40 revenue per order

MTailor

March 2022 – July 2022

Contractor · US Remote

- Executed complex backend migration from Parse/MongoDB to **Firebase Firestore** and GCP infrastructure
- Built migration scripts to read data from MongoDB and write to Firestore after running complex processing logic
- Created technical documentation for engineering team on new Firebase SDKs (Node.js and Python)
- Trained Operations team on new Firebase Dashboard and backup procedures

Dr Wealth

August 2021 – January 2022

Software Engineer · Singapore Remote

- Optimized PWA delivering real-time stock market prices from Firebase backend
- Built ad-hoc jobs to query over 2 million Firestore records, keeping customer prices current from Morningstar APIs
- Extended backend APIs using Firebase Cloud Functions and Express.js hosted on Heroku

CodeBits

January 2020 – July 2021

Software Engineer · Uganda Remote

- Led team of 4 developers building legal tech systems for NGOs in Uganda
- Designed and deployed **containerized microservices infrastructure** using **Docker, Kubernetes, and Apache Kafka** for event-driven communication at FIDA Uganda
- Built USSD service for legal aid providers communicating with Node.js backend via **gRPC**
- Developed cross-platform mobile app for LASPNET using React Native and Expo with Firebase push notifications

KEY PROJECTS

Parse → Firebase Cloud Migration MTailor

Architected and executed zero-downtime migration of 20+ production applications from AWS/Parse stack to GCP/Firebase. Built real-time two-way sync pipeline using Pub/Sub to maintain data consistency during parallel operation of both systems.

Node.js, Python, Google Pub/Sub, Firebase, MongoDB, AWS S3, GCS

Event-Driven Microservices Platform CodeBits

Designed microservices architecture for FIDA Uganda's case management system. Implemented Apache Kafka for inter-service event streaming, containerized with Docker and orchestrated via Kubernetes.

Apache Kafka, Docker, Kubernetes, Node.js, gRPC, NATS Streaming

Real-Time Stock Price System Dr Wealth

Built data pipeline to sync 2M+ Firestore records with Morningstar APIs, delivering real-time stock prices to PWA users across multiple markets.

Firebase Cloud Functions, Firestore, Express.js, React

EDUCATION

Bachelor of Science in Computer Engineering — Makerere University

2018 – 2022

Kampala, Uganda

TECHNICAL SKILLS

Cloud: AWS (S3, EC2, EBS), GCP (Firebase, Pub/Sub, Cloud Functions, GCS) **Containers:** Docker, Kubernetes

Messaging: Apache Kafka, NATS Streaming, gRPC, Google Pub/Sub **Languages:** Node.js, Python, Go, JavaScript, TypeScript

Databases: MongoDB, Firestore, SQL **Frontend:** React, React Native, HTML/CSS, Tailwind